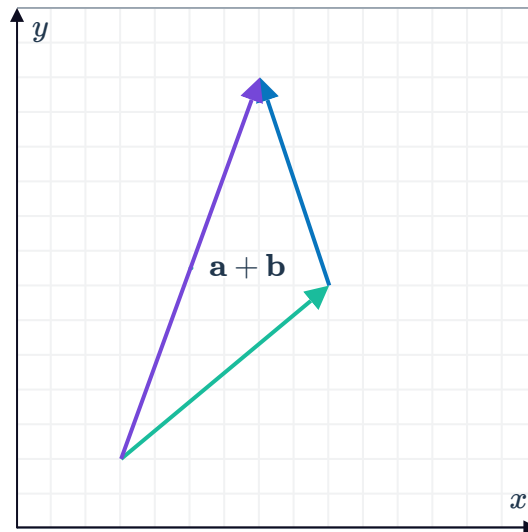


Geometric addition and subtraction ectors (Extended)

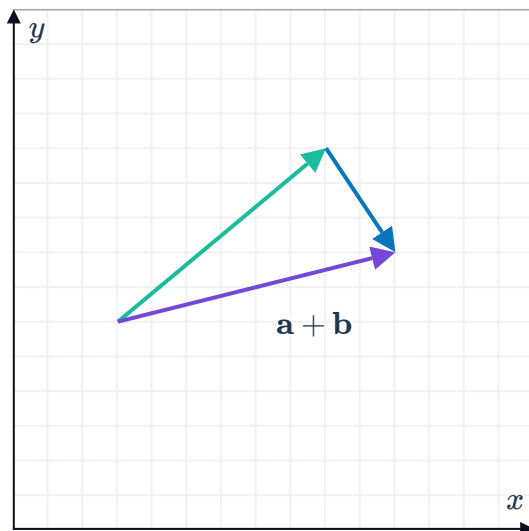
1 c

2

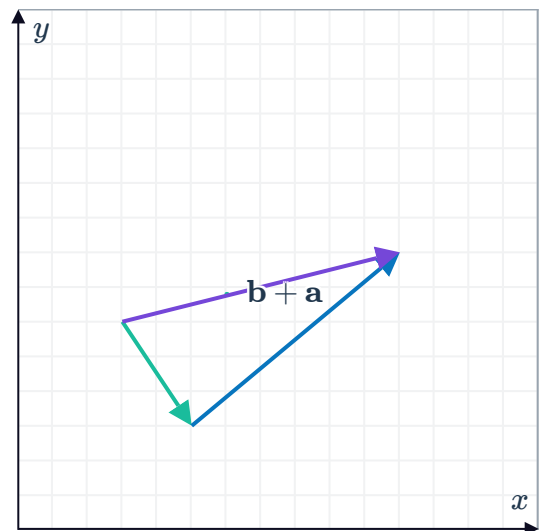


3

a

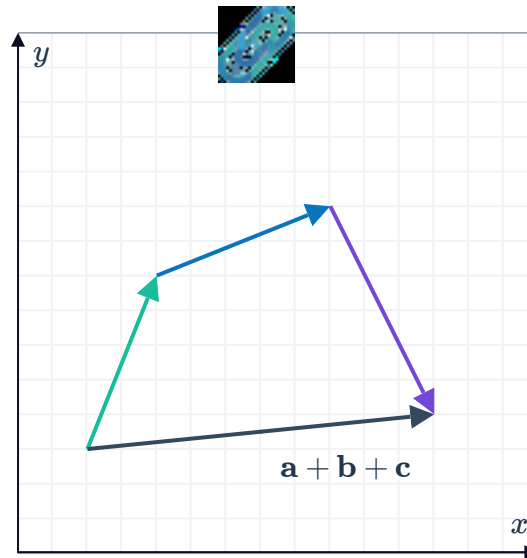


b

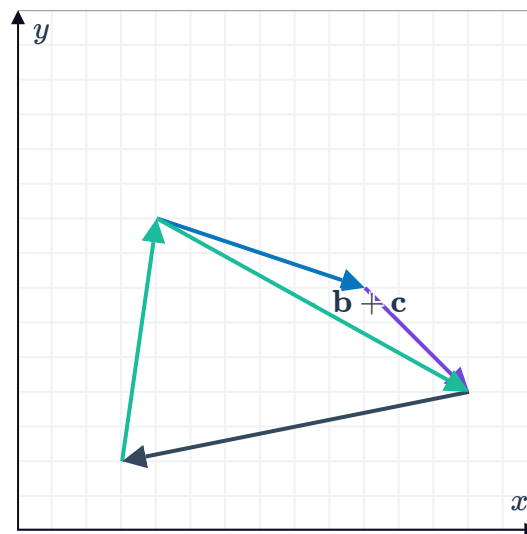


c Yes

4



5



Algebraic addition and subtraction of vectors

6 $(13, 1)$

7

a

$$\text{i} \begin{pmatrix} 12 \\ 11 \end{pmatrix} \quad \text{ii} \begin{pmatrix} 15 \\ 7 \end{pmatrix} \quad \text{iii} \begin{pmatrix} 19 \\ 16 \end{pmatrix}$$

b

$$\text{i} \begin{pmatrix} -10 \\ -1 \end{pmatrix} \quad \text{ii} \begin{pmatrix} -2 \\ -7 \end{pmatrix} \quad \text{iii} \begin{pmatrix} -8 \\ 0 \end{pmatrix}$$

c

$$\text{i} \begin{pmatrix} 12.1 \\ -3.9 \end{pmatrix} \quad \text{ii} \begin{pmatrix} 2.3 \\ -1.2 \end{pmatrix} \quad \text{iii} \begin{pmatrix} 7 \\ -7.4 \end{pmatrix}$$

d

$$\text{i} \begin{pmatrix} \frac{13}{30} \\ -\frac{1}{5} \end{pmatrix} \quad \text{ii} \begin{pmatrix} \frac{11}{10} \\ -\frac{11}{5} \end{pmatrix} \quad \text{iii} \begin{pmatrix} \frac{29}{15} \\ -1 \end{pmatrix}$$



Scalar multiplication of vectors

8 Let $\mathbf{a} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$, find:

a $2\mathbf{a}$

b $5\mathbf{a}$

c $-1\mathbf{a}$

d $-6\mathbf{a}$

9 Let $\mathbf{a} = \begin{pmatrix} 10 \\ 16 \end{pmatrix}$, find:

a $3\mathbf{a}$

b $-10\mathbf{a}$

c $\frac{1}{2}\mathbf{a}$

d $-\frac{1}{4}\mathbf{a}$

10 Let $\mathbf{a} = \begin{pmatrix} -4 \\ 9 \end{pmatrix}$, find:

a $3\mathbf{a}$

b $-2\mathbf{a}$

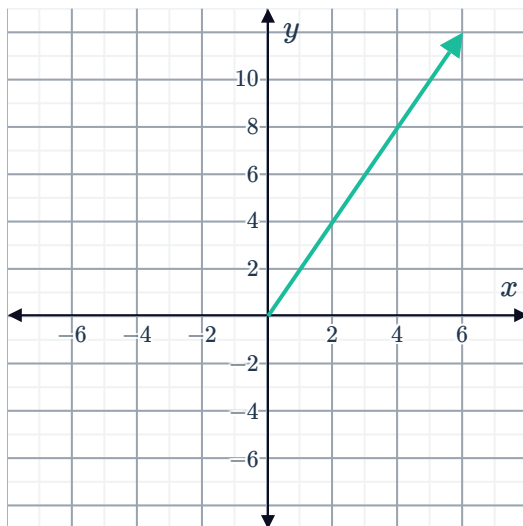
c $\frac{1}{2}\mathbf{a}$

d $-100\mathbf{a}$

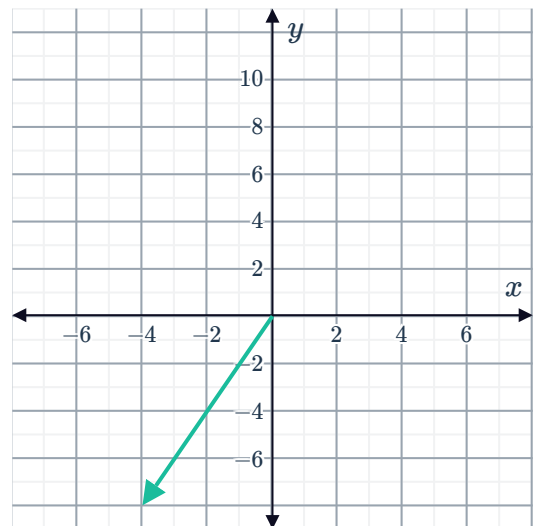
Scalar multiplication of vectors (Extended)

11

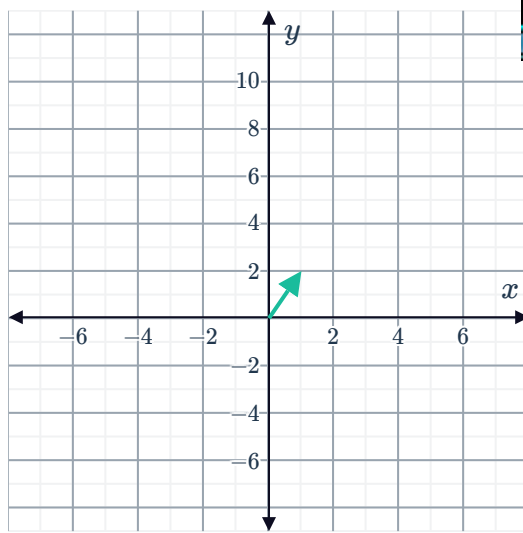
a



b



c



12

a $\begin{pmatrix} -1 \\ 11 \end{pmatrix}$

b $\begin{pmatrix} 9 \\ 15 \end{pmatrix}$

c $\begin{pmatrix} -16 \\ 24 \end{pmatrix}$

d $\begin{pmatrix} 19 \\ 19 \end{pmatrix}$

e $\begin{pmatrix} -24 \\ -2 \end{pmatrix}$

13

a b

b c

c d

d c

14

a $\begin{pmatrix} 6 \\ 6 \end{pmatrix}$

b $\begin{pmatrix} -12 \\ 16 \end{pmatrix}$

c $\begin{pmatrix} -6 \\ -6 \end{pmatrix}$

d $\begin{pmatrix} 1 \\ 4 \end{pmatrix}$

e $\begin{pmatrix} 11 \\ 9 \end{pmatrix}$

f $\begin{pmatrix} 14 \\ 20 \end{pmatrix}$

15 $x = 1, -1$

16 $x = 1$